

Milind Kumar Mishra

Product Engineer

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Experience

(3y 10m total)

Product Engineer / Design Engineer

FULL TIME

Thine / Merlin AI by Foyer

Feb 2025 — Present · 1y 3m

Own frontend execution across Merlin's customer-facing product surfaces and Thine's web experiences, spanning product engineering, UX systems, payments, and AI-native interfaces.

- Owned product experience, user experience, and frontend implementation for a browser agent built inside a Chrome extension sidebar
- Led design system improvements across the Merlin website to raise UI consistency, implementation speed, and overall product quality
- Maintained various payment flows for the Merlin website, ensuring a smoother purchase and subscription experience for customers and better campaign outcomes for the marketing team
- Owned the frontend for Merlin's customer-facing web product, delivering core user journeys across merlin projects, engagement, retention and support experiences
- Built interaction-heavy AI product flows for Merlin, including imports, chat history improvements, merlin projects, and responsive motion systems
- Developed the Thine website and shaped its product storytelling through a polished, image sequence scroll experience

[Thine's Website](#) · [Merlin AI's Website](#) · [Merlin's Chrome Extension](#)

Software Engineer

FULL TIME

SARAL - The Influencer OS

Dec 2024 — Feb 2025 · 2m

Improved internal product workflows for support, growth, and campaign operations.

- Revamped internal dashboard to improve visibility of key metrics for support and growth teams
- Implemented multi-select drag-and-drop interactions for faster bulk operations
- Built a content submission UI to streamline approvals and accelerate campaign execution

[Saral's Website](#)

Founding Product Engineer

FULL TIME

Proof-of-Skill protocol

Jun 2024 — Dec 2024 · 6m

Led product engineering for an assessment and skill validation platform from MVP to deployment.

- Led frontend development for MVP flows across Skill Validators, Candidates, and Recruiters
- Designed and implemented validator UI with voting and consensus-driven assessment validation
- Built real-time assessment recording and proctoring integrations to improve data integrity and mal-practice detection
- Architected core systems and handled end-to-end deployment for platform availability

[Proof of Skill Protocol's Website](#)

Lead Frontend Engineer

StartupHire

FULL TIME

Aug 2023 — Jan 2024 · 5m

Led frontend execution for recruiting workflow and Gen AI product prototypes.

- Led a team of two developers to build a recruiting pipeline prototype
- Integrated multiple job boards to centralize candidate applications and improve recruiter visibility
- Managed deployments for Gen AI workflows to ensure stable operation

Frontend Engineer / Research Assistant

Locus Connect / National Yang Ming Chiao Tung University

FULL TIME

Jul 2022 — Jul 2023 · 1y

Built product, research, and visualization systems for a 3D indoor positioning platform across industry and university collaboration.

- Developed and maintained frontend for a 3D indoor positioning product and related marketing experiences
- Maintained frontend systems for indoor positioning with integrated IoT data pipelines
- Enhanced 3D visualization using React, GSAP, and Three.js
- Improved HDOP calculations and z-index accuracy from around 20 cm to under 10 cm
- Managed infrastructure tasks across switch configuration, load balancers, Docker deployments, and bare-metal services

[Locus Connect's Website](#) [University Website](#)

Selected Projects

Expense Agent

Apr 2026

Full-stack AI bookkeeping agent with an Express API, Next.js 16 PWA, Telegram bot, and MCP server that tracks expenses, income, and savings through natural conversation with receipt vision, voice memos, and Gmail extraction

ROLE Full Stack Product Engineer **TEAM** 1 **DURATION** Apr 2026 - Present **STATUS** Active

- Architected and built the entire stack solo, from the PostgreSQL schema and Express REST API to the Next.js 16 PWA frontend, Telegram bot, and MCP server, all wired together as a Bun + Turbo monorepo
- Designed a unified ledger in PostgreSQL with Drizzle ORM where one transactions table powers expenses, income, and savings, with multi-currency FX conversion resolved at query time and migrations managed through Drizzle Kit
- Built three independent transports (REST endpoints, Telegram bot tools, and MCP tools) over a single shared domain layer so business logic stays consistent across every surface
- Implemented conversational ingestion with AI SDK 6 ToolLoopAgent, including image-to-expense extraction from receipts via vision models, voice memo transcription via Whisper, and PDF and spreadsheet preprocessing
- Integrated Composio-backed Gmail tooling so the agent can read inbox receipts and extract structured transactions on demand, scoped per user via Better Auth identity
- Shipped the Next.js 16 PWA frontend with iOS install support, same-origin proxy rewrites for Safari compatibility, persistent chat history, and a streaming chat UI built on the AI SDK
- Wired OpenTelemetry plus Langfuse instrumentation into the agent loop and tool calls for full prompt, token, and latency observability across every transport
- Auto-generated OpenAPI types from express-zod-api v27 endpoints and shared them across API and frontend through a typed workspace package, eliminating manual API type drift
- Built a self-serve data export pipeline backed by S3-compatible storage with signed download URLs and a pluggable collector registry for new user-owned tables

[API Docs](#) [Website](#)

JSON Visualiser

Oct 2024

JSON inspection and visualization tool for debugging deeply nested payloads across tree and grid views

ROLE Product Engineer **TEAM** 1 **DURATION** Oct 2024 - Present **STATUS** Live

UNIQUE USERS 143 **RETURN RATE** 43%

- Built a unified inspection workflow that combines validation, formatting, and multi-view exploration for complex payloads
- Implemented tree and grid renderers optimized for nested JSON structures with efficient expand and collapse interactions
- Reached 143 unique users with a 43% return rate, including one user with 49 sessions as of Mar 2026

[Live](#) [Source](#)

Markdown Visualizer

Dec 2025

Markdown editing and preview tool for fast iteration on technical writing workflows

ROLE Product Engineer TEAM 1 DURATION Dec 2025 - Present STATUS Live

- Built a low-latency editing and preview workflow to reduce context switching while drafting mark-down-heavy content
- Implemented rendering and editing utilities that keep technical writing workflows responsive and predictable

Live Source

Talks

Building a Component Distribution System with shadcn Registry	React Bangalore
Mastering ViewTransition in React for Stunning UI Updates	React Play Bengaluru
Building Real-Time Applications with Reactive Databases	React Play x React Bangalore
AI for React Developers	React Bangalore

Open Source Contributions

feat: add table markdown copy and csv/markdown download options #99	vercel/streamdown
feat: add download functionality to code blocks #102	vercel/streamdown
feat: add image download functionality with hover controls #103	vercel/streamdown
feat: add speech to text input to prompt area and update test and example apps #112	vercel/ai-elements

Education

National Yang Ming Chiao Tung University · Research Program, CSE	Jul 2023
Visvesvaraya Technological University · Bachelor of Engineering, ECE	Sep 2022

Skills

JavaScript	TypeScript	React	Next.js	Node.js	Tailwind CSS	HTML	CSS	Git	Docker	REST APIs
MQTT	Zustand	React Query	Vercel AI SDK	OpenAI API	D3.js	Framer Motion	Figma	CI/CD		